Roots introduce low internal arguments

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Abstract

This paper argues that Roots introduce their own internal arguments (IAs) as complements. Korean non-derived nominals with argument-licensing capacity (Yoon & Park 2008) and their light verb (LV) predicate counterparts must utilize a $\sqrt{}$(Root)P (Harley 2014), as both verbal and nominal functional alternatives are empirically inadequate. Comparison across different types of LV constructions reveals that low IAs introduced directly by Roots and higher IAs introduced by verbal functional structure can be empirically differentiated by differential object marking, with low IAs necessarily originating in an acategorical constituent (the $\sqrt{}$P). Consequently, Roots are fully functioning syntactic heads that project phrases and take complements, contrary to the mainstream view of Roots in decompositional morphosyntactic theory, which assumes Roots to be syntactically deficient, unable to introduce complements and unable to exist in a structure without assistance from categorizing heads (e.g., Alexiadou 2014; Borer 2003, 2014; Lohndal 2020; Merchant 2019; Embick & Marantz 2008).

Keywords

[syntax] [morphology] [argument structure] [Roots] [categorization]

1 The √P Hypothesis

This paper addresses crucial questions on the nature of Roots[1] within syntactic theory—can Roots live a syntactic life? Can they introduce internal arguments? What would an empirical argument in favor of these commitments look like?

Harley (2014) argues in the affirmative: Roots must be individuated as formal syntactic units, to which semantic and phonological properties can then be accessed at a later point in the derivation. Her grounds for this proposal come from phonological and semantic suppletive paradigms: on the one hand, Semitic templatic Roots (1) have a consistent phonological shape, but a wide variety of semantic uses, proving it difficult to pinpoint a single articulated semantic definition for the Root itself. On the other hand, Hiaki unaccusative Roots (2) exemplify the reverse paradigm, in which a Root with one reliable meaning has full phonological suppletion, realizing two different forms depending on object number agreement. Hiaki suppletive verbs make it difficult to pinpoint a single underlying abstract phonological form for the Root itself. Given that neither phonological nor semantic individuation proves sufficient for differentiation between Roots, Harley (2014) posits individuation on the basis of pure syntactic units of structural computation.

(1) Semitic Root √*sgr* with some of its idiosyncratic interpretations

a.	CaCaC (v)	*sagar*	v, 'close'
b.	hitCaCCeC (v)	*histager*	v, 'cocoon oneself'
c.	CoCCayim (n)	*sograyim*	n, 'parentheses'
d.	miCCeCet (n)	*misgeret*	n, 'frame'

(Harley 2014: 264)

(2) Some Hiaki unaccusative verbs with full suppletion paradigms

a.	*vuite* ˜ *tenne*	'run.sg ˜ run.pl'
b.	*siika* ˜ *saka*	'go.sg ˜ go.pl'
c.	*weama* ˜ *rehte*	'wander.sg ˜ wander.pl'
d.	*kivake* ˜ *kiime*	'enter.sg ˜ enter.pl'

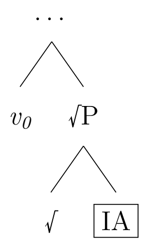
(Harley 2014: 234)

For Harley (2014), the Hiaki suppletive verb paradigms (2) also provide empirical motivation for treating Roots as the introducers of internal arguments (IAs). Assuming suppletion to be a local operation (Bobaljik 2012), it must be the case that the IA, which conditions the suppletive forms in (2), exists in a local (sisterhood) relationship with the Root, such that the suppletive form is then correctly selected during Vocabulary Insertion (Anderson

[1] As in the formal category-neutral units assumed in decompositional morphosyntactic theory (e.g., Pesetsky 1995; Marantz 1997; Borer 2003; Embick & Marantz 2008).

1992; Aronoff 1993; Beard 1995; Bobaljik 2008, 2012; Embick & Marantz 2008). The Harley (2014) proposed structure for Hiaki verbs therefore utilizes the √P, exemplified in (3). In what follows, I refer to this structural proposal as the *√P Hypothesis*.

(3) The √P Hypothesis



Most contemporary theories of predicate structure, however, have a different viewpoint, arguing that Roots play no role in structural generation (Borer 2003, 2005a,b, 2013; Lohndal 2014) and cannot be direct introducers of complements (Alexiadou 2014; Merchant 2019), even requiring adjunction to a categorizing head in order to enter the syntactic derivation (Embick 2004; Embick & Marantz 2008; Embick 2015; Lohndal 2020). IAs must then, under this perspective, instead be introduced by functional projections, not by Roots, since Roots lack the ability to project phrases. In what follows, I refer to this perspective as the *Deficient Root Hypothesis*. Most *Deficient Root Hypothesis* approaches to argument structure (AS) assert that verbal functional projections, specifically, are the introducers of thematic arguments (Borer 2013; Alexiadou & Lohndal 2017a,b; De Belder 2011; Lohndal 2014; Riksem 2018), largely driven on data from derived nominal paradigms in different languages. Only a restricted set of nominal phrases cross-linguistically can license thematic arguments[2], and the ones that do (AS-Nominals; Grimshaw 1990), have been found in many languages to grammatically license VP adverbs and VP anaphora, indicating that their AS properties correlate with independent syntactic evidence for verbality (Hazout 1995; Fu et al. 2001; Alexiadou 2001, 2009, 2010a,b). Restricting argument introduction to verbal functional heads therefore effectively constrains the grammar such that only nominals containing embedded verbal structure will exhibit properties of AS.

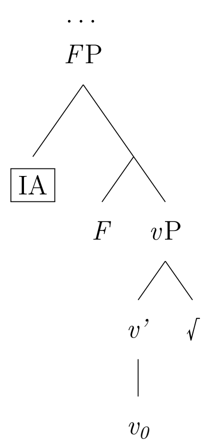
The *√P Hypothesis* stands in direct opposition to the *Deficient Root Hypothesis*: if Roots can introduce their own IAs as their complements, then they are syntactic heads, and consequently are not 'syntactically deficient' elements that need assistance from other heads to enter a derivation. Proving the *√P Hypothesis* empirically, however, comes down to being able to isolate and justify the existence of an acategorial syntactic constituent that consists solely of the Root and its internal argument. Harley (2014) argues for this constituent on

[2]Specifically, event participants. See Barker & Dowty (1993) for an independent proposal about non-eventive thematic roles in nominals.

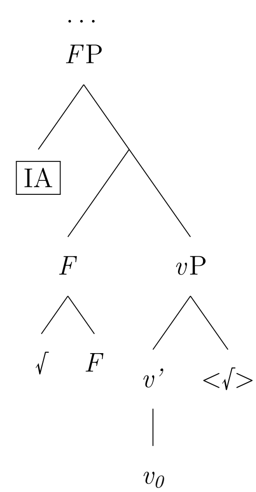
the basis of suppletive paradigms, but, as Alexiadou (2014) points out, this argument is contingent on the identification of specific Spell-Out domains. Since Harley (2014) suggests that the conditioning environment for the Root suppletion must be up to VoiceP, given data on idioms, it is not strictly necessary for the IA to originate as a complement of the Root—it could be plausible that the IA is introduced higher, by some functional projection *F*, and a copy of the Root moves into *F* via head movement, at which point it would enter into a sufficiently local relationship with the IA to trigger suppletion (4).

(4) Alternative structure with *F* (Alexiadou 2014; Alexiadou & Lohndal 2017b)

a. *F* introduces IA



b. Move √ to F



Thus, unless it can independently be proven that phonological conditioning and semantic conditioning are sensitive to independent Spell-Out Domains (phonological conditioning at the level of the √P or the categorizing head v_0, but semantic conditioning at Voice), suppletive verbs in Hiaki do not provide a strong enough empirical argument to disprove the *Deficient Root Hypothesis.*

The primary contribution of the present paper is an empirical argument in favor of the *√P Hypothesis* that crucially makes no reference to Spell-Out domains. Rather, I arrive at the necessity of the √P through the decoupling of AS and syntactic categorization. The *Deficient Root Hypothesis* requires all argument-introducing elements to be verbal functional heads. If thematic arguments can be found licensed in a completely non-verbal structure, then we can conclude that the *Deficient Root Hypothesis* as it stands is inadequate to properly capture the empirical landscape. Given that non-verbal AS-nominals have indeed been identified in Korean (Yoon & Park 2008), I show that it must be the case that these AS-nominals have a √P structure, in order to correctly capture different types of light verb constructions in

Korean, whose structures directly depend on whether they are built from Roots that do or do not introduce their own IA. Consequently, Roots are *not* syntactically deficient units that require assistance from categorizing heads, but rather are fully functioning formal syntactic heads that project and take their own complements.

Section 2 characterizes the long-standing debate on whether AS involves verbal structure or not, and summarizes the upshots of Yoon & Park (2008), who demonstrate that there are AS-nominals in Korean which categorically lack verbal structure, forcing the conclusion that AS can exist in without verbal functional projections. Section 3 argues that it must be the Roots at the core of the non-verbal Korean AS-Nominals responsible for introducing the IA, first by ruling out all nominal functional projection alternatives, then through a comparison of AS-Roots and non-AS Roots that have the same syntactic distribution, yet license IAs differently depending on the Root involved. Section 4 provides the formal analysis, and Section 5 contends with the double case marking phenomenon in Korean (e.g., Choe 1987; Cho 2000; Chae 1997; Ko 2007; Sim 2004; Ura 1996; Yoon 1990, 1991, 2015), providing further independent evidence for the *√P Hypothesis* via an extension of the account to part-whole nominals. Section 6 concludes.

Unless otherwise stated, Korean data examples presented in this manuscript come from my own data collection, in conversation with five separate native Korean language consultants, with additional verification from a number of native Korean audience members who were present for various presentations of this work in its earlier stages.

2 The relationship between AS and verbal structure

The history of work on derived nominals, also referred to as nominalizations, spans decades of syntactic research (See Rozwadowska 2017 for a comprehensive overview). At its core, this investigation has aimed to find an answer to whether, when examining a nominal that exhibits properties of AS (henceforth AS-Nominal), if these properties arise from the presence of verbal structural layers embedded inside of them or not. AS-Nominals have been central to the Lexicalist vs. Structuralist debate for precisely this reason: if AS is constructed in the Lexicon (e.g., utilizing a theta-grid), then a noun like *destruction* inherits its arguments from an associated verb *destroy* within the Lexicon, but nonetheless generates a purely nominal structure (Aronoff 1976; Chomsky 1970; Giorgi & Longobardi 1991; Grimshaw 1990; Jackendoff 1975; Lieber 1980). If AS is constructed in the Syntax, however, then the structural elements that add arguments into a VP must be embedded inside any nominal structure that also demonstrates AS properties (Alexiadou 2001; Alexiadou & Rathert 2010; Borer 2003, 2005a,b, 2013; Engelhardt 1998; Fu et al. 2001; Hazout 1995; Lohndal 2014, 2020).

Contemporary theories, thanks to major contributions from predicate decomposition work (e.g., Hale & Keyser 1993; Harley 1995; Kratzer 1996; Larson 1988; Pesetsky 1995), largely fit into the Structuralist perspective, saying yes, AS-Nominals have AS properties due to embedded VP structure. There are three main reasons for this: First, with more granular decompositional machinery, we now have the tools needed to introduce arguments using functional projections, crucially without requiring a transformational approach that converts full sentences to nominals (the main issue at hand in Chomsky 1970). Second, there are independently identified empirical issues with Lexicalism (e.g., Mirror Principle; Baker 1985, 1988), that have lead to the development of morphosyntactic frameworks looking to do word formation syntactically and severely limit the power of the Lexicon (Halle & Marantz 1993; Lieber 1992), making it conceptually desirable to move AS out of the Lexicon and into the Syntax, as well. Third, it has been found consistently in a number of languages, including Hebrew, Greek, and English, that VP adverbs and VP anaphora are licensed in AS-Nominals, even when the nominal itself has an unambiguously nominal morphological form (i.e., surfacing with affixes that unambiguously mark them as nominal, thereby distinguishing it from a gerund). For example, both Alexiadou (2001) and Hazout (1995) use evidence of adverbial modifiers within nominalizations in Greek and Hebrew respectively to argue for a VP layer inside of an AS-Nominal. Fu et al. (2001) report the same for English, showing that English speakers find the presence of VP adverbs to be acceptable in AS-Nominals (5), but not in non-AS nominals (6).

(5) a. His explanation of the accident thoroughly (did not help him).
  b. His transformation into a werewolf so rapidly was unnerving.

(6) a. * His version of the accident thoroughly (did not help him).
  b. ?? His metamorphosis into a werewolf so rapidly was unnerving.

(Fu et al. 2001: 555)

Also useful is discussion from Alexiadou (2009), which notes that the result/process ambiguity (Grimshaw 1990) that many English derived nominals exhibit, does not exist for nominals that have overt verbalizing morphology. The result/process ambiguity refers to derived nominals like *assignment* having two possible interpretations: one as an AS-Nominal, and one as a simple referential nominal (7).

(7) a. The frequent assignment of unsolvable problems frustrated the students.
  b. The student forgot to turn in their mathematics assignment.

However, if the surface form of the nominalization unambiguously indicates the presence of a verbalizing functional head, such as the morpheme *-ize* in *verbalization* (8), the AS reading is obligatory. Alexiadou (2009) therefore provides the same generalization as Fu et al. (2001): if a construction has verbal functional structure, then it necessarily has AS.

(8) The verbalization of their concerns took a long time/*was on the table. (Alexiadou 2009: 254)

One way this generalization has been captured is in terms of *phrasal coherency*, which theorizes about the entailment relationships between VP properties (Borer 2003; Malouf 1998; Yoon & Park 2008). Sentential cases (e.g., nominative/accusative) are included in the set of structural properties associated with the VP, as a property that also entails the presence of thematic arguments (and therefore AS). Verbal structural properties entailing AS sets a strong expectation for co-occurrence such that AS and verbal structure will go hand in hand. However, to prove the perspective under the *Deficient Root Hypothesis*, that AS is *exclusively* licensed by verbal functional structure, this entailment relationship must be symmetrical: Not only must verbal structure entail AS, AS must also entail verbal structure.

Yoon & Park (2008) falsify this prediction. They deal with AS-Nominals in Korean built from a class of lexical items often referred to as Verbal Nouns (VNs) in the Korean and Japanese syntax literature. The term *Verbal Noun*, similar to the terms *nominalization* and *derived nominal*, reflects a contentious back-and-forth about the correct category label and derivational history of these lexical items (Ahn 1992; Chae 1996, 1997; Jun 2003, 2006; Kim et al. 2007; Manning 1993; Pak 2001; Park 2013; Sells 1995; Yoon & Park 2008). Despite this shared history with derived nominals/nominalizations in Western linguistic literature, Yoon & Park (2008) critically show that, unlike AS-Nominals in English, Greek, and Hebrew, AS-Nominals built from Korean VNs categorically disallow verbal structural properties. They provide an example ruling out VP adverbials (9); (10–11) provide additional data points.

(9) *mikwun-uy* *baghdad-ulo-uy* *sinsok-**han/*-hi*** *cinkyek*
American.army-GEN Baghdad-to-GEN quick-ADJ/*-ADV incursion
"American troops' quick incursion into Baghdad" (Yoon & Park 2008: 235)

(10) [*ainsyuthain-uy* *ppal-**un/*-li*** *pich-uy* *sokto(-uy)* *kyeysan*]-*un* ...
[Einstein-GEN quick-ADJ/*-ADV light-GEN speed(-GEN) calculation]-TOP
"Einstein's quick calculation of the speed of light (... was impressive/etc.)"

(11) [*yenkwuwen-uy* *kkunhimeps-**nun/*-i*** *tongkwul* *thamkwu*]-*nun* ...
[researcher-GEN constant-ADJ/*-ADV cave explore]-TOP
"The researcher's constant exploration of the cave (... was tiring/etc.)"

Additionally noted by Yoon & Park (2008), genitive case (GEN) is the only felicitous case marking available within AS-Nominals headed by VNs. Sentential cases nominative (NOM) and accusative (ACC) are impossible (12).

(12) [*yenkwuwen-**uy**/*-**i*** *kkunhimeps-nun* *tongkwul*(***-uy*/*-*ul***) *thamkwu*]-*nun*
[researcher-GEN/*-NOM constant-ADJ cave(-GEN/*-ACC) explore]-TOP
"The researcher's constant exploration of the cave (... was tiring/etc.)"

Yoon & Park (2008) further confirm that these AS-Nominals, despite lacking the ability to license VP adverbials and sentential case marking, still satisfy all of the longstanding diagnostics for AS from Grimshaw (1990), which target the event structure (ES) (Aktionsart; Vendler 1957) of the process denoted by the phrasal head. For example, (13) demonstrates licensing of agent-oriented modifiers, and (14) demonstrates licensing of temporal modifiers.

(13) a. *cheli-uy* ***uytoceki-n*** *yenghi-eytayha-n* *pinan*
Cheli-GEN intentional-ADJ Yenghi-be.about-ADJ criticism
"Cheli's intentional criticism of Yenghi" (Yoon & Park 2008: 235)

b. *sacang-uy* ***kyeyhwoekceki-n*** *cungke(-uy)* *unphyey*
boss-GEN deliberate-ADJ evidence(-GEN) conceal
"the boss' deliberate concealment of evidence"

(14) a. *mikwun-uy* ***sam.il*** ***tongan-uy*** *Baghdad(-uy)* *kongkyek*
American.army-GEN three.days during-GEN baghdad(-GEN) attack
"America's attack on Baghdad for three days" (Yoon & Park 2008: 235)

b. *yenkwuwen-uy* ***i.nyen*** ***tongan-uy*** *tongkwul(-uy)* *thamkwu*
researcher-GEN two.year duration-GEN cave(-GEN) explore
"the researcher's exploration of the cave for two years"

Korean AS-Nominals headed by VNs also license event modifiers, e.g., *frequent* and *constant* (15), and manner adjectives, such as *slow* or *quick* (16).

(15) a. *cikwen-uy* ***cac-un*** *kongkum(-uy)* *hoynglyeng*
worker-GEN frequent-ADJ fund(-GEN) embezzle
"the worker's frequent embezzlement of funds"

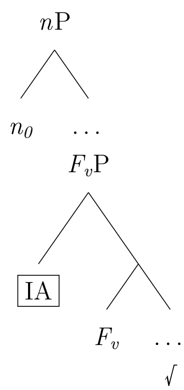
b. *yenkwuwen-uy* ***kkunhimeps-nun*** *tongkwul(-uy)* *thamkwu*
researcher-GEN constant-ADJ cave(-GEN) explore
"the researcher's constant exploration of the cave"

(16) *ainsyuthain-uy* ***ppal-un/nuli-n*** *pich-uy* *sokto(-uy)* *kyeysan*
Einstein-GEN quick-ADJ/slow-ADJ light-GEN speed(-GEN) calculate
"Einstein's quick/slow calculation of the speed of light"

Korean nominals built from VNs, then, are indeed argument-bearing nominals (per Grimshaw 1990's diagnostics), but fail to license any properties associated with verbal structure. These AS-Nominals are fully non-verbal, and provide definitive proof that AS can exist in the absence of verbal structure.

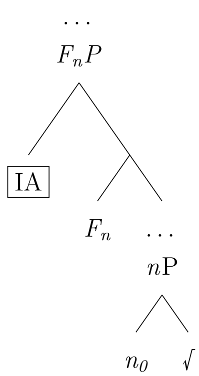
Within a framework that assumes Roots, the specific structure ruled out by Yoon & Park (2008)'s contribution is one in which an internal argument[3] of a Korean AS-Nominal is introduced by a verbal functional projection, as in (17).

(17) Ruled out structure for Korean AS-Nominals



Crucially, what is impossible is an IA-introducing verbal projection embedded inside of the nominal. In attempting to, as much as possible, maintain the worldview of the *Deficient Root Hypothesis*, one might instead look to interpret the Korean facts from Yoon & Park (2008) as a unique phenomenon of Korean VNs, where *nominal* functional structure instead accomplishes argument introduction, e.g., (18).

(18) Nominal functional head introduces IA



Expanding the empirical landscape to consider the VNs and their argument realization in sentential environments will give us the necessary data to prove that the nominal functional structure alternative sketched in (18) is untenable. As such, I next turn to Korean light verb

[3]Given the focus on √P in this paper, I here set licensing of external arguments (EAs) aside, as Roots contend with the lowest/first-introduced argument, the IA. I assume that EAs are divorced from the predicate and, in the verbal domain, licensed separately by Voice (Kratzer 1996). For the nominal domain, I adopt the viewpoint that external arguments in AS-nominals, derived or otherwise, are not introduced in the same manner as EAs in the verbal domain. See Sichel (2010) for a useful discussion on the similarities and differences of EA licensing, among other event structure (ES) properties, in VPs vs. nominalizations.

constructions (henceforth LVCs), which famously involve heavy use of VNs in both Korean and Japanese[4] (Ahn 1992; Chae 1996, 1997; Grimshaw & Mester 1988; Jun 2003, 2006; J. R. Kim 1993; J. B. Kim et al. 2007; S. W. Kim 1994; Manning 1993; Miyamoto & Kishimoto 2016; Pak 2001; Park 1995; Sells 1995; J. H. S. Yoon 1990; J. M. Yoon 1997).

3 Advocating for argument-introducing Roots

If verbal structure does not drive AS in the Korean nominals, then what does? I will argue that it is the Root. Yoon & Park (2008) data on AS-nominals in Korean reveal that AS-nominals in Korean fully lack verbal structure, and yet pass diagnostics for having AS. This means that the AS of these constructions cannot have a verbal source. Given this result, the label "VN" is truly a misnomer for these lexical items, although I will continue to use this term as needed so as to remain in direct conversation with the broader literature on these items. I will argue in this section that what sets VNs apart from other lexical items is that they are, in fact, pronunciations of argument-taking Roots (AS-Roots).

3.1 Nominal functional structure is insufficient

There are two immediate alternatives to consider in the face of these Korean AS-Nominals: one, of course, is the *√P Hypothesis*, but, as already stated at the end of Section 2, proponents of the *Deficient Root Hypothesis* could counter with the alternative proposal that, for just these lexical items in Korean, their AS is instead uniquely created through the use of nominal structure (18). If the IA of these constructions is licensed by a nominal functional projection, and not by the Root, then, in other argument-licensing contexts where the same Roots are used, nominal structure should be diagnosable. Just as VP adverbials and sentential cases are reliable diagnostics for verbal functional structure, adjectival modifiers and nominal cases (i.e., genitive) are available as diagnostics for nominal functional structure.

VNs are used extremely productively in LVC environments— in fact, almost all literature on Korean VNs, sans Yoon & Park (2008) is primarily interested in investigating VNs and the means by which they interact with light verbs to create complex predicate structures. Some basic examples of VNs in LVCs are provided in (19) below.

(19) a. *ainsyuthain-i pich-uy sokto-lul* ***kyeysan-ha****-yss-eyo*
Einstein-NOM light-GEN speed-ACC calculate-do-PST-DECL
"Einstein calculated the speed of light."

[4] Comparable phenomena are found in a number of languages beyond Korean and Japanese as well, including Mongolian, Hindi/Urdu, Persian, K'iche', Chuj, and Zapotec (Butt 1995, 2010; Can Pixabaj & Aissen 2021; Mahajan 2012; Megerdoomian 2012; Mohanan 2017; Özbek 2010).

b. *cikwen-i* *kongkum-ul* ***hoynglyeng-ha**-yss-eyo*
worker-NOM fund-ACC embezzle-do-PST-DECL
"The worker embezzled the funds."

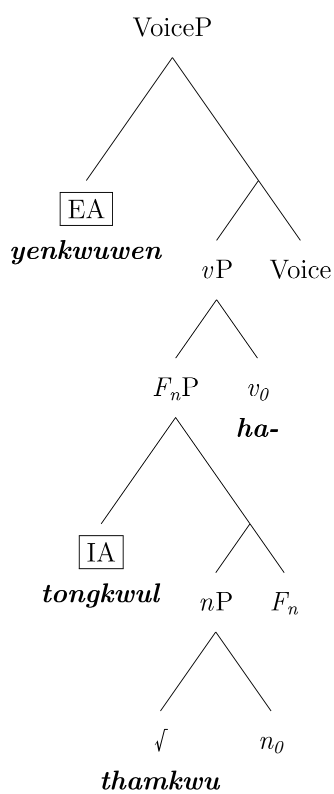
c. *yenkwuwen-i* *tongkwul-ul* ***thamkwu-ha**-yss-eyo*
researcher-NOM cave-ACC explore-do-PST-DECL
"The researcher explored the cave."

Unsurprisingly given the fully non-verbal nature of Korean AS-Nominals made by VNs, a light verb component is required in order to generate a grammatical sentential construction with a VN (20), which follows straightforwardly from the understanding that VNs are, on their own, incapable of generating verbal structure.

(20) * *ainsyuthain-i* *pich-ey* *sokto-lul* ***kyeysan**-yss-eyo*
Einstein-NOM light-of speed-ACC calculate-PST-DECL
Intended: "Einstein calculated the speed of light."

Though a light verb effectively licenses a VN within a verbal structure,it cannot be the presence of the light verb itself that creates the correct conditions for licensing of the IA, since the same IA is independently license-able in a fully nominal structure (Yoon & Park 2008). For LVCs like (19), then, proponents of the *Deficient Root Hypothesis* need to assume the structure (21), where a nominal functional projection F_n introduces the IA.

(21) Possible AS for Korean LVC: IA is introduced by F_n



This structure effectively treats the LVC as a noun incorporation construction, where the light verb is a verbalizing head that takes the AS-Nominal as its complement, and the AS-Nominal is then incorporated into the light verb[5]. Noun incorporation is, in fact, a common analysis of LVCs like (19) (see, e.g., Grimshaw & Mester 1988).

However, the proposal in (21) commits to underlying nominal functional structure, which is not borne out empirically. In an LVC, adjectival modification of the VN is fully impossible (22), and so is genitive case (23).

(22) *yenkwuwen-i* *tongkwul-ul* *kkunhimeps-**i/*-nun*** *thamkwu-ha-yss-eyo*
researcher-NOM cave-ACC constant-ADV/*-ADJ explore-do-PST-DECL
"The researcher tirelessly/continuously explored the cave."

(23) *yenkwuwen-**i/*-uy*** *tongkwul(**-ul/*-uy**)* *thamkwu-ha-yss-eyo*
researcher-NOM/*-GEN cave(-ACC/*-GEN) explore-do-PST-DECL
"The researcher explored the cave."

The inability to license adjectival modifiers and genitive case, properties of the nominal domain, means that nominal functional structure is absent from Korean LVCs.

Further evidence against (21) is that the light verb (LV) *ha-* in Korean does not select for nominals, but rather, selects for Roots directly. Though *ha-* selects for many lexical items that live separate lives as nominals, it can also select for items that, crucially, *must* be Roots. This is most transparent in mimetic/onomatopoetic descriptive predicates, such as the examples in (24–26). Mimetic Roots can combine with the LV *ha-* to create descriptive predicates, as well as with nominalizing suffixes to create referential nouns.

(24) *mallang(mallang)*, the sound/notion of squishi-ness
a. *mallang(mallang)-ha-* $\sqrt{}$*SQUISHY*-do- "to be squishy"
b. *mallang-i* $\sqrt{}$*SQUISHY*-NMLZ.DIM "(a) Squishy"

(25) *yaong(yaong)*, the "meow" sound that a cat makes
a. *yaong(yaong)-ha-* $\sqrt{}$*MEOW*-do- "to meow"
b. *yaong-i* $\sqrt{}$*MEOW*-NMLZ.DIM "(a) kitty cat"

(26) *kaykol(kaykol)*, the "ribbet" sound that a frog makes
a. *kaykol(kaykol)-ha-* $\sqrt{}$*CROAK*-do- "to croak/ribbet"
b. *kaykol-i* $\sqrt{}$*CROAK*-NMLZ.DIM "a frog"

Crucially, mimetic Roots cannot live independent lives as nominals on their own, even if the intended referent of the nominal is the sound/notion itself (27).

[5] Though, the structure in (21) is conceptually at odds with typical treatments of noun incorporation, which often assume the incorporated element to be structurally simple (e.g., just of a category N).

(27) a. * *(kaykwuli-tul-uy) kaykwulkaykwul-i tul-li-yss-eyo*
(frog-PL-GEN) croak.croak-NOM hear-PASS-PST-DECL
Intended: "A (frogs') croak was heard."

b. *(kaykwuli-tul-i) kaykwulkaykwul-ha-nun soli-ka tul-li-yss-eyo*
(frog-PL-NOM) croak.croak-do-ADJ sound-NOM hear-PASS-PST-DECL
"The sound of frogs croaking was heard."

In sum, given the absence of nominal structure (22–23), and additional evidence that *ha-* selects for items that are unambiguously Roots (24–27), there are no empirical grounds to insist on the presence of any nominal layers embedded within the argument structure of Korean LVCs involving VNs.

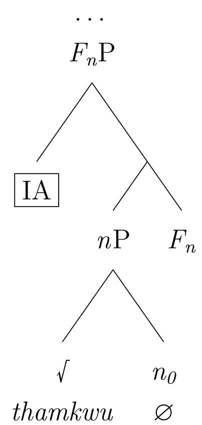
None of this rules out the open possibility for head movement of the Root to the node exponed by the LV, which, in a certain sense, is not unlike the characterization of noun incorporation as head movement (Baker 1988, 2009; Hale & Keyser 1993; Travis 1984), but, crucially, does not require a nominal category. Conceptually, the idea of the LV *ha-* composing with a √(P) directly, therefore lending it an incorporation-like characterization, also provides an explanation for why noun incorporation is so often characterized as involving a 'bare noun' or nominal that is lacking structure in some way (Johns 2017; see also Johns 2007 for a treatment of some instances of incorporation as an LV combining with a √P).

In Korean AS-Nominals and LVCs, neither nominal or verbal structure can be responsible for introducing the internal argument. If the necessary property for AS is not verbal, nor is it nominal, what is left? The only thing consistent across these constructions is the Root, i.e. the VN itself. Therefore, it is the Root that creates the possible conditions for AS.

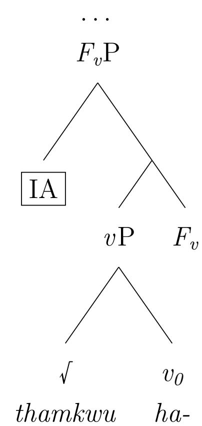
3.2 Only some Roots introduce IAs

A theory of AS that restricts argument introduction to only verbal functional projections is inadequate, and treating Korean VNs as exceptional lexical items that exclusively utilize a nominal functional projection for IA-introduction is also inadequate. Defendants of the *Deficient Root Hypothesis*, in face of these outcomes, would have no choice but to next suggest two different functional projections for introducing an IA, one that is a nominal flavor, and one that is a verbal flavor, and that both are used with structures involving VNs (28). The nominal functional head would be used in crafting the AS-Nominal, and the verbal functional head would be used in crafting the LVC. Under this account, the only difference between the two structures is that in one, the Root was adjoined to a fully nominal, argument-introducing structure, and in the other, to a fully verbal, argument-introducing structure.

(28) a. F_n introduces IA (AS-Nominal)



b. F_v introduces IA in (LVC)



This immediately creates an over-generation problem, especially under a fully Exoskelatal approach where the Root has no ability at all in constraining the projections above it (e.g., Borer 2005a,b, 2013). If there is both a nominal means and a verbal means to include an IA, there is no reason why we shouldn't expect all types of nominals to be capable of AS. This is, of course, not borne out: it is already well-established that the nominal domain is cross-linguistically more constrained in its argument-hosting capacities than the verbal domain (Abney 1987; Alexiadou 2001; Grimshaw 1990; Pesetsky 1995; Sichel 2010).

There is a clear minimal comparison immediately available within Korean that rules out (28): VNs can be directly contrasted with another set of lexical items that have the same environmental distribution, but are unable to introduce their own arguments, further showing that it is not the structural environment that a Root surfaces in which determines whether AS is available, but the Root itself. For example, consider the LVCs in (29). In each, the LV *ha-* combines with an item (*mal* and *nolay*, respectively) that can independently live a life as a nominal (30), the same as VNs. These LVCs also, importantly, can take a direct object.

(29) a. *cwuni-ka mwuncang-ul khu-key* ***mal***-*ha-yss-eyo*
Juni-NOM sentence-ACC big-ADV word-do-PST-DECL
"Juni said (the) sentence loudly."

b. *ku kaswu-ka cayen-uy alumtawum-ul cacwu* ***nolay***-*ha-yss-eyo*
that singer-NOM nature-GEN beauty-ACC frequently song-do-PST-DECL
"That singer often sang of nature's beauty."

(30) a. *i mwuncang-un* ***mal***-*i manh-ayo*
this sentence-TOP word-NOM be.many-PRS.DECL
"This sentence has a lot of words."

b. *hanyengi-ka* *kacang* *cohaha-nun* ***nolay**-ka* *latio-eyse* *hullenao-ass-eyo*
Hanyoung-NOM most like-ADJ song-NOM radio-on flow.out-PST-DECL
"The song (that) Hanyoung likes most played on the radio."

Crucially, however, lexical items like *mal* and *nolay* cannot create AS-Nominals (31). Therefore, though an ACC-marked object is possible in the LVC constructions (29), the source of this internal argument cannot be the Root, since it is only licensed when verbal structure is present.

(31) a. * *Cwuni-uy* *cac-un* *mwuncang(-uy)* ***mal***
Juni-GEN frequent-ADJ sentence(-GEN) word
Intended: "Juni's frequent saying of sentence(s)"

b. * *kaswu-uy* *cac-un* *cayen-uy* *alumtawum* ***nolay***
singer-GEN frequent-ADJ nature-GEN beauty song
Intended: "The singer's frequent singing of nature's beauty"

Unlike the VN constructions, we have no choice but to posit a verbal functional projection that introduces the IA for these lexical items— crucially, despite the total overlap in the structural environments that both VNs and these other items occur in (LVCs, nominals). This minimal pair reveals that the IA licensed by a VN is unique to the Root of that VN: In other words, VNs have their AS properties because they are pronunciations of argument-introducing Roots (AS-Roots). Items like *mal* and *nolay*, on the other hand, are pronunciations of Roots that lack argument-introducing properties (non-AS Roots).

Intriguingly, the examples in (29) reveal that the verbal domain has alternatives means through which an IA may be added to the structure, thereby allowing for direct objects to still be licensed by complex predicates created from non-AS Roots. No such means exists within the nominal domain, hence the ungrammaticality of (31). We therefore arrive at the observation that not all internal arguments are introduced by the same means: there are two types of IAs, ones that are introduced by Roots, and ones that are introduced by a verbal functional projection. Beyond being different with regard to the respective domain they are associated with, these two types of IAs are also predicted to differ in their structural height: specifically, IAs introduced by verbal functional projections should be structurally *higher* that IAs introduced by Roots. Differential object marking (DOM; Bossong 1985, 1991; Aissen 2003) provides us the means to prove that these two IA base positions are diagnosable and structurally distinct. Though Korean does not have DOM fully grammaticalized as a categorical distinction between objects (in contrast to languages like Turkish, e.g. Enç 1991; Öztürk 2005), DOM has still been identified and investigated in Korean by a number of researchers (Chung 2020; Kim 2008; Ko 2000; Kwon & Zribi-Hertz

2008; Lee 2005, 2006a,b, 2015). There are two possible canonical positions for the internal argument to appear: either directly adjacent to the head of the relevant phrase (32), or higher, above modification (adverbial or adjectival, respectively) (33).

(32) a.

cikwen-i	*cacwu*	***kongkum(-ul)***	*hoynglyeng-ha-yss-eyo*
worker-NOM	frequently	fund(-ACC)	embezzle-do-PST-DECL

"The worker frequently embezzled (the) funds."

b.

cikwen-uy	*cac-un*	***kongkum(-uy)***	*hoynglyeng*
worker-GEN	frequent-ADJ	fund(-GEN)	embezzle

"the worker's frequent embezzlement of (the) funds"

(33) a.

cikwen-i	***kongkum*(-ul)***	*cacwu*	*hoynglyeng-ha-yss-eyo*
worker-NOM	fund*(-ACC)	frequently	embezzle-do-PST-DECL

"The worker frequently embezzled the funds."

b.

cikwen-uy	***kongkum*(-uy)***	*cac-un*	*hoynglyeng*
worker-GEN	fund*(-GEN)	frequent-ADJ	embezzle

"the worker's frequent embezzlement of the funds"

Optionality of case marking on the IA is determined by position: when directly adjacent to the phrase head (32), case is optional, and can readily be dropped. If, however, the IA precedes a modifier (33), case is then much harder to drop: without strong inherent notions of definiteness, specificity and/or animacy, case on the IA in this position is obligatory[6].

Importantly, the overlap in domains reveals that, not only do both domains have the same case mechanism for an IA, but also that the base position of the IA across both of the domains must be the same, such that it interacts with case assignment in the exact same way. The modifier position also serves to diagnose the positions with respect to height: the position preceding the modifier is a structurally higher one, where the IA is unambiguously licensed within the verbal or nominal domain, respectively. The position directly adjacent to the phrasal head, maps the IA to a low position that is exists lower in the structure than *any domain-specific structural elements*. In other words, the low position (32) is complement to the Root. The optionality of case in (32) follows from modifier attachment ambiguity: when the IA is case marked, it is unambiguously licensed within the verbal domain (32a) or nominal domain (32b), and adjuncts may attach either above or below it in the structure. When bare, we are dealing with an IA that is structurally lower than the modifier, i.e. one that is not licensed within a specific domain, but rather by the Root itself.

[6] In the sentential domain, some speakers find it possible to elide the accusative case marker in the higher object position (e.g., 33a) given an IA that is highly animate, definite, unique, specific, etc. For example, a proper name as a direct object is more easily amenable to case drop in this position. This phenomenon is still distinct from the behavior of case marking in the predicate adjacent position (e.g., 32a), where case drop is always possible.

As optional case in the low IA position is tied to being the complement of a Root, it follows that optional case should disallowed in LVCs built from non-AS Roots, since these Roots can never take a complement. This is exactly what we find when we investigate accusative case in non-AS Root LVCs. Accusative case is obligatory on the IA regardless of whether it follows (34) or precedes (35) the adverbial modifier[7].

(34) a. *Cwuni-ka* *khu-key* ***mwuncang*(-ul)*** *mal-ha-yss-ta*
Juni-NOM big-ADV sentence*(-ACC) word-do-PST-DECL
"Juni said (the/a) sentence loudly."

b. *kaswu-ka* *cacwu* ***cayen-uy*** ***alumtawum*(-ul)*** *nolay-ha-yss-eyo*
singer-NOM frequently nature-GEN beauty*(-ACC) sing-do-PST-DECL
"The singer often sang of nature's beauty."

(35) a. *Cwuni-ka* ***mwuncang*(-ul)*** *khu-key* *mal-ha-yss-eyo*
Juni-NOM sentence*(-ACC) big-ADV word-do-PST-DECL
"Juni said (the/a) sentence loudly."

b. *kaswu-ka* ***cayen-uy*** ***alumtawum*(-ul)*** *cacwu* *nolay-ha-yss-eyo*
singer-NOM nature-GEN beauty*(-ACC) frequently sing-do-PST-DECL
"The singer often sang of nature's beauty."

The obligatory-ness of case for the IA regardless of position, when looking at non-AS Root LVCs, follows from the fact that the IA for these constructions is only licensed by a verbal functional projection: the lower position, complement to a Root, is not possible, and as such the IA cannot grammatically be bare. Regardless of the formal case assignment theory adopted, we must conclude from these data that the structures built from AS-Roots have more space: the internal argument originates as a complement to the Root, and can optionally surface higher[8]. This low position is unavailable for non-AS Roots.

4 Formalization

In this section, I provide formal structural proposals for AS-Nominals in Korean built from AS-Roots, complex predicates in Korean built from AS-Roots, and complex predicates in Korean built from non-AS Roots. The analysis adopts a morphosyntactic framework in line

[7]DOM in the nominal domain for non-AS Roots is, of course, not testable, since non-AS Roots cannot create AS-Nomminal constructions.

[8]It is not necessary to assume that the higher position the IA appears in for AS-Root constructions, is necessarily the same position where the IA is introduced for non-AS Root constructions. It is possible for the landing site for marked direct objects to be a distinct position somewhere near the edge of the full verbal phrase/nominal phrase, and also possible to assume— for theory coherence purposes— that an IA introduced by a verbal functional projection could still undergo object shift as well, to this high position.

with Distributed Morphology (Halle & Marantz 1993), assuming that pronounced morphological form of the syntactic structure is handled by a post-syntactic module. Borrowing terminology from Ramchand (2008), I take the relevant thematic 'role' of an IA introduced directly by an AS-Root to be that of an UNDERGOER. Neo-Davidsonian event semantics is adopted (Davidson 1967; Parsons 1990), and I also assume, following the insights of predicate decomposition research, that all arguments added by verbal functional projections combine with the predicate via Event Identification (36) (Kratzer 1996; Higginbotham 1985; see also implementations in Doron 2003; Pylkkänen 2008; among others).

(36) Event Identification:
$\textsc{Ident}(\alpha_{\langle e,\langle s,t\rangle\rangle}, \beta_{\langle s,t\rangle}) \equiv \lambda P.\lambda y_e.\lambda e_s.[\alpha(e,y) \wedge P(e)](\beta)$

Section 4.1 provides example argument structures for AS and non-AS Roots, with corresponding compositional semantics. Section 4.2 provides the assumed syntactic structures, and contends with the realization of the light verb *ha-* in LVCs built from both types of Roots. I analyze *ha-* as an exponent of a sequence of verbal functional projections, rather than assuming it to simply be a pronunciation of the categorizing head v_0.

4.1 Predicate structure of AS and non-AS Roots

The investigation of Korean AS-Nominals, LVCs, and the Roots at their core, have revealed that low internal arguments are introduced by Roots, not by functional projections. This result requires the adoption of the *√P Hypothesis*. Roots cannot be syntactically deficient elements as the *Deficient Root Hypothesis* posits, but rather, Roots are fully fledged syntactic heads that can project and take their own complements. However, given that Roots can differ in their AS capabilities, Roots must also encode some amount of semantic information, that is accessible immediately upon entering the syntactic derivation. This characterization departs from the Harley (2014) analysis, where she suggests that Roots do not have semantic individuation. Given the existence of both AS and non-AS Roots, a Root must, minimally, provide the information of whether or not it has an internal argument. I propose to understand this formally with entailment: if the event (or entity) denoted by the Root necessarily entails the existence of a thematic participant, then that participant may be realized directly as the complement of the Root itself.

For example, the Korean AS-Root *√EXPLORE* entails an UNDERGOER (the entity that undergoes exploration), and as such has the following semantic denotation in (37). *√EXPLORE* combines directly with its IA in the first step of structural composition (38).

(37) $[\![explore]\!] = \lambda x_e.\lambda e_s.[explore(x_e)(e_s)]$

(38) $\sqrt{}$P composition for of $\sqrt{EXPLORE}$

- $\sqrt{}$P: $\lambda e_s.[explore(x_e)(e_s)]$
  - $\sqrt{}$: $\lambda x_e.\lambda e_s.[explore(x_e)(e_s)]$
  - DP: x_e

The structure in (38) serves as the foundation for both the AS-Nominal and the VP built from $\sqrt{EXPLORE}$, depending on whether a nominal or verbal functional layer is next added into the tree.

A non-AS Roots does not entail the presence of a thematic participant, and consequently no such participant may be realized as its complement. For example, the Korean non-AS Root $\sqrt{WORD}$ does not entail any thematic participants, and has the denotation in (39).

(39) $[\![word]\!] = \lambda x_e.[word(x_e)]$

In constructing the non-AS Root LVC, we must coerce the Root $\sqrt{WORD}$ from type ⟨e, t⟩ to a predicate of type ⟨s, t⟩. I adopt below in (40) a formalization from Doron (2003), positing first a verbal head F_{v_1} that adds an event argument. Then, a second verbal projection F_{v_2} introduces an IA via the process of Event Identification (Kratzer 1996). I use the thematic role label Theme, here, to distinguish the high IA from the low IA.

(40) Predicate structure built from $\sqrt{WORD}$

- F_{v_2}P: $\lambda e_s.[Theme(y_e)(e_s) \wedge word(e_s)]$
  - DP: y_e
  - $\lambda x_e.\lambda e_s.[Theme(x_e)(e_s) \wedge word(e_s)]$
    - F_{v_2}: $\lambda x_e.\lambda e_s.Theme(x_e)(e_s)$
    - F_{v_1}P: $\lambda e_s.[word(e_s)]$
      - F_{v_1}: $\lambda P_{\langle e,t\rangle}.\lambda e_s.[P(e_s)]$
      - $\sqrt{}$: $\lambda x_e.[word(x_e)]$

The broader morphological distribution of the light verb *ha-* in Korean, which we turn to next, requires that these two functional projections are represented separately, as opposed to assuming one F_v that accomplishes both IA introduction and transformation to predicate.

4.2 Handling the light verb *ha-*

The light verb *ha-* in Korean plays a role in allowing both AS-Roots and non-AS Roots to exist in verbal structures. This characterization immediately lends itself to the possibility that *ha-* is a categorizing head, i.e., v_0. Under the *Deficient Root Hypothesis*, categorizing heads like v_0, n_0, a_0, etc. serve the function of licensing Roots in a structure (Embick & Marantz 2008); the findings here, however, in favor of the *√P Hypothesis*, beg the question whether v_0 is in fact a strictly necessary head to posit in the structure of a Korean LVC at all. If Roots are fully fledged syntactic heads on their own, then they don't require categorizers to license them. In light of this, one could instead, for example, adopt a Borerian approach to categorization (Borer 2014), where category is determined through association with a particular structural domain built above the Root, rather than linked to the presence or absence of a syntactic head that serves the sole function of categorization.

The empirical distribution of *ha-* is complicated: in some cases, such as in LVCs built with AS-Roots, it really does seem to do little more than mark the presence of the verbal domain (a characterization in favor of treating it as simply a categorizer). On the other hand, *ha-* in LVCs built with non-AS Roots seems to do a lot more heavy lifting, surfacing in a structure that has added properties of AS that the Root in isolation lacked. Even sorting LVCs by AS and non-AS Roots, though, does not fully capture a systematic patterning of *ha-* and its function. For AS Roots, *ha-* adds no AS properties to the structure, as both the event argument and IA are licensed by the Root. Nonetheless, the larger set of light verbs available in Korean (e.g. *twoy-*, *sikhi-*, *pat-*; English: "become", "order", "receive") reveal that the choice of LV form is still sensitive to *something* in the underlying predicate structure. For example, for AS-Root LVCs, there are reliable *ha-*/*twoy-* ("do"/"become") valency alternations, e.g. (41), where *ha-* reliably surfaces for the transitive form, while *twoy-* reliably surfaces for the unaccusative form.

(41) a. *yenkwuwen-i tungkwul-ul thamkwu-**ha**-yss-eyo*
researcher-NOM cave-ACC explore-do-PST-DECL
"The researcher explored the cave."

b. *tungkwul-i thamkwu-**twoy**-yss-eyo*
cave-NOM explore-become-PST-DECL
"The cave was/got/became explored."

For non-AS Roots like *mal* and *nolay*, the use of *ha-* correlates with the addition of both an event argument and IA. However, there are many more non-AS Root LVC predicates with *ha-* that are, in fact, intransitive (42). These predicates involve Roots that can independently head referential nominals (though crucially not AS-Nominals). Given that there are these intransitive uses of *ha-* involving non-AS Roots, in addition to the handful that add a direct object, it is not possible to analyze *ha-* as a single head that provides both the event argument and the IA. In (42), the functional head F_{v_1} that transforms the ⟨e, t⟩ Root to an ⟨s, t⟩ predicate is present, but the functional head F_{v_2} that adds an IA is not.

(42) Some example intransitive predicates that surface with the light verb *ha-*:

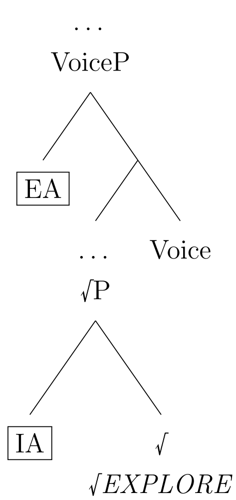
a.	*samang-ha-*	√*DEATH*-do-	"to die"
b.	*sanchayk-ha-*	√*STROLL*-do	"to take a walk"
c.	*il-ha-*	√*THING*-do-	"to work"
d.	*swukcey-ha-*	√*HOMEWORK*-do-	"to do homework"

In sum, for AS-Roots, *ha-* does not add either event structure (ES) nor an IA, both of which are contributed by the Root itself— but, the choice of *ha-* over other LVs suggests that its form is not completely arbitrary. For non-AS Roots, *ha-* correlates with the addition of ES, but not necessarily with the addition of a high IA. The distribution of *ha-*, therefore, does not correlate neatly with one distinct functional head of the predicate structure.

We can understand the messiness of this empirical landscape as an indication that *ha-* as a morphological form does not correspond to the same underlying structure in all instances. I propose to treat *ha-* as a pronunciation of the verbal functional sequence (up to Voice) that is utilized as a PF repair strategy when the Lexicon only has a stored Vocabulary Item for the Root itself. This account utilizes the mechanism of *Spanning* (Svenonius 2016; Bye & Svenonius 2012), an account which asserts that "spans", rather than terminal nodes, are the loci for lexical insertion in a post-syntactic approach to morphological form. A span is defined as a contiguous sequence of heads in a head-complement relation.

Take, for example, the LVC *thamkwu-ha-* ("to explore"), to have the structure in (43).

(43)



I suggest that the AS-Roots explored in this paper have forms stored in the lexicon as direct exponents of the Root. For *thamkwu*, this would be as in (44). Crucially, because a corresponding form to expone the full span of the verbal predicate does not exist, a light verb is chosen as a repair, to pronounce the span of heads above the Root.

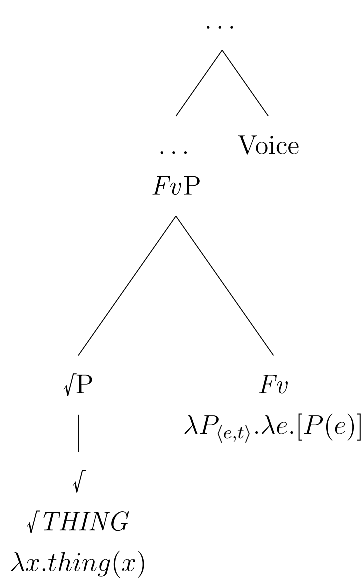
(44) Lexical entries

a. *thamkwu*, /t$^{\mathrm{h}}$am.ku/ $\Leftrightarrow$ $\langle\, \surd EXPLORE \,\rangle$

b. !(DOES NOT EXIST) $\Leftrightarrow$ $\langle\, \surd EXPLORE, \ldots, \text{Voice}_{(\text{ACTIVE})} \,\rangle$

This same analysis straightforwardly derives the intransitive predicates from (42) as well. Take, for example, the intransitive predicate *il-ha-* ("to work"): I assume an underlying structure like (45), which has at least two verbal heads above the Root: the functional projection F_{v_1} that "eventivizes", and Voice.

(45)



(46) Lexical entries

a. *il*, /il/ $\Leftrightarrow$ $\langle\, \surd THING \,\rangle$

b. !(DOES NOT EXIST) $\Leftrightarrow$ $\langle\, \surd THING, \ldots, Fv, \ldots, \text{Voice}_{(\text{ACTIVE})} \,\rangle$

Again, because there is only a stored entry for the Root (46), rather than the full span of the predicate structure, the light verb *ha-* is inserted to pronounce the remaining portion of the span.

This analysis treats the choice of light verb form as dependent on the sequence that it pronounces. I assume that *ha-*, minimally, is the default. It seems plausible that the choice of LV is sensitive to Voice (e.g., *ha-* as active Voice, *twoy-* possibly as a middle Voice), but I leave the full treatment of which heads condition the LV forms beyond *ha-* to future research.

5 Double case as independent evidence for AS-Roots

This section addresses a separate phenomenon in Korean syntactic research: double case marking (Choe 1987; Cho 2000; Chae 1997; Ko 2007; Sim 2004; Ura 1996; Yoon 1990, 1991, 2015). Double accusative case marking has been used to explicitly argue that VNs license ACC case (Chae 1996, 1997), which, under the frameworks used in this paper, would require VNs to contain verbal structure. I briefly summarize and rule out the concerns raised by Chae (1996, 1997, 2003) about the lexical category of VNs, and then argue that the distribution of double case marking in Korean in fact provides further evidence in favor of the √P Hypothesis.

LVCs involving VNs have an alternative construction where *ha-* acts as a main verb, rather than a light verb, in which case the VN projects an AS-Nominal that surfaces as the direct object of *ha-* (47). While (47) is entirely expected, the surprising variant that has drawn the attention of many researchers is (48), where the IA of the VN also receives ACC case. It is possible to separate the ACC-marked IA from the ACC-marked VN with a sentential adjunct (49).

(47) *yenkwuwen-i* *[tongkwul(-uy)* *thamkwu]-lul* *kkunhimeps-i* *ha-yss-eyo*
researcher-NOM [cave(-GEN) explore]-ACC constant-ADV do-PST-DECL
"The researcher continuously explored the cave."

(48) *yenkwuwen-i* *[tongkwul]-ul* *[thamkwu]-ul* *kkunhimeps-i* *ha-yss-eyo*
researcher-NOM [cave]-ACC [explore]-ACC constant-ADV do-PST-DECL
"The researcher continuously explored the cave."

(49) *yenkwuwen-i* *[tongkwul]-ul* *ecey-nun* *[thamkwu]-ul* *ha-yss-eyo*
researcher-NOM [cave]-ACC yesterday-TOP [explore]-ACC do-PST-DECL
"The researcher explored the cave yesterday."

Chae (1996, 1997) proposes that in these double accusative constructions, as well as in "aspectual nominal constructions" (50), it is the VN itself that licenses ACC case.

(50) *tongkul(-uy/-ul)* *thamkwu* *cwung*
cave(-GEN/-ACC) explore middle
"during exploration of the cave/while exploring the cave ..."

The item *cwung* literally means "center", and when combined with a predicate, it picks out a temporal viewpoint "during" or "while" the corresponding event is taking place. VNs can occur directly adjacent to *cwung*, and, unlike the Korean AS-nominal constructions, here both nominal and sentential case markers are possible.

Crucially, however, the presence of ACC in both of these environments necessarily co-occurs with functional structure that selects for the constituent containing the VN and its IA (either a light verb, or *cwung*). As per discussion in Section 2, AS-Nominals headed by VNs disallow sentential cases (Yoon & Park 2008), and as such, the option for ACC case licensing on the IA in (48–49) and (50) must come from something beyond the VN, i.e. from verbal structure. Double case is also possible for part-whole nominals (51) (Choe 1987; Cho 2000; Ura 1996; Yoon 1990, 1991, 2015); under Chae (1996, 1997)'s analysis, we'd have to assume that "hand" assigns ACC case to "Hanyoung" in (51b).

(51) a. *Nikho-ka* [*Hanyengi(-uy)* *son*]-*ul* *cap-ass-eyo*
Niko-NOM [Hanyoung(-GEN) hand]-ACC grab-PST-DECL
"Niko grabbed Hanyoung's hand."

b. *Nikho-ka* [*Hanyengi*]-*lul* [***son***]-*ul* *cap-ass-eyo*
Niko-NOM [Hanyoung]-ACC [hand]-ACC grab-PST-DECL
"Niko grabbed Hanyoung's hand."

Among the analyses of Korean double case for part-whole nominals, some support a constituent analysis where the "whole" originates as a complement to the "part" (Choe 1987; Ura 1996; Cho 2000; Ko 2007). Others argue for a non-constituent approach, where both are arguments of the verbal predicate, introduced separately (Kim 1989; Yoon 1990; Sim 2004). Importantly, VNs and part-whole nominals are the only phenomena that allow double case in Korean (Yoon 1997). Alienable possession constructions, for example, disallow double case (52). So do non-AS Root LVCs[9] (53).

(52) a. *Nikho-ka* [*Hanyengi-uy* ***kapang***]-*ul* *cap-ass-eyo*
Niko-NOM [Hanyoung-GEN bag]-ACC grab-PST-DECL
"Niko grabbed Hanyoung's bag."

b. * *Nikho-ka* [*Henyengi*]-*lul* [***kapang***]-*ul* *cap-ass-eyo*
Niko-NOM [Hanyoung]-ACC [bag]-ACC grab-PST-DECL
Intended: "Niko grabbed Hanyoung's bag."

(53) a. * *Cwuni-ka* [*mwuncang(-uy)* ***mal***]-*ul* *(khu-key)* *ha-yss-eyo*
Juni-NOM [sentence(-GEN) word]-ACC (big-ADV) do-PST-DECL
Intended: "Juni said the sentence (loudly)."

[9]Some native Korean-speaking colleagues have noted to me that there is a possibility for (53b) to be acceptable under a contrastive focus reading of the second accusative case marker, on *mal*. The use of Korean case and discourse markers as indicators of contrastive focus is a known phenomenon (see Jun 2015 and citations therein) but is not yet well understood formally. What is crucial here is that, even if there is a reading under which (53b) is acceptable, it cannot be treated as the same double case phenomenon observed for VNs and part-whole constructions.

b. */? *Cwuni-ka* [*mwuncang*]-*ul* [***mal***]-*ul* *(khu-key)* *ha-yss-eyo*
Juni-NOM [sentence]-ACC [word]-ACC (big-ADV) do-PST-DECL
Intended: "Juni said the sentence (loudly)."

Part-whole nominals are cross-linguistically observed to have unique properties setting them apart from other types of possessive constructions (Barker & Dowty 1993; Boneh & Sichel 2010). Boneh & Sichel (2010), in an investigation of possession constructions in Palestinian Arabic, argue for heterogeneous representation of different possessive constructions in the language. Of particular relevance to the discussion here, they argue that the part-whole relation is the only one to have a head-complement source: the "whole" must be introduced as a complement to the head that denotes the "part" (under the current framework, as a complement to the Root).

When it comes to double case marking in Korean, regardless of whether we ultimately adopt a constituent analysis involving movement, or a non-constituent analysis (perhaps involving delayed argument saturation), the only types of constructions that allow the double case phenomenon to begin with are ones that, crucially, independently allow a √P structure. Just as the Root √*EXPLORE* entails an entity (an undergoer event participant), the Root √*HAND* entails an entity as well: namely, the "whole".

6 Conclusion

This paper argues in favor of the *√P Hypothesis*, through an investigation of Korean LVCs and the Roots at their core, in order to empirically disentangle the licensing of an internal argument from verbal functional structure. I motivate the existence of the √P constituent in a series of steps: first, I rule out both verbal and nominal IA-introducing projections for structures built from AS-Roots in Korean. Then, I demonstrate that these AS-Roots have different IAs from non-AS Roots, despite both types of Roots being licensed in the same environments (LVCs and nominals). Finally, differential object marking serves to diagnose the two different IA positions, proving that the low position exists as complement to the Root, prior to the addition of domain-specific functional structure.

The paper also argues explicitly against the *Deficient Root Hypothesis*, which is incompatible with the characterization of Roots under the *√P Hypothesis*. Given that Roots syntactically project and introduce their own complements, they are real syntactic heads, just as any other head in the structure, and do not need categorizers to license them.

The proposal predicts that AS is always non-nominal, but not exclusively verbal. There are no nominal functional projections that introduce thematic arguments, and any IA that we find in a nominal construction exhibiting only nominal syntactic properties (i.e., a non-derived AS-Nominal) is an IA that was introduced by the Root itself. In the verbal domain, additional functional layers introduce all other arguments, including high internal arguments (such as the high IAs of non-AS Root LVCs), and all external arguments (e.g. Beneficiaries, Agents, Causers: Pylkkänen 2008; Kratzer 1996; Alexiadou et al. 2015).

Roots introducing IAs, though incompatible with a strictly Exoskeletal approach (e.g. Borer 2005a,b, 2013), need not be a full return to a Lexicalist framework. The spirit of this conclusion, rather, can be thought of as a way of incorporating longstanding Lexicalist generalizations into a non-Lexicalist framework. The proposed structures for the Korean AS-Nominal and LVC constructions respectively are remarkably identical in shape, despite having no derivational relationship: they share the same $\sqrt{}$P base and only differ in the type of functional domain that is built above them. The resulting fully parallel nominal and verbal structures are consistent with the original X-bar theoretic vision (Chomsky 1970), without needing to throw out all of the gains that have been made in decomposing the structure of the VP. In a decompositional view of a predicate, we still allow for structure to play a large and meaningful role in the licensing of AS: the challenge becomes defining exactly what it is that the Root can constrain, and what is dictated by the functional structure. I've argued here that the Root drives the introduction of a low internal argument, and that all other argument types are licensed by verbal functional structure.

Acknowledgments

Thank you to Ivy Sichel, for her consistent support and guidance at every stage of this project: I owe much of the intellectual development of this research to her expertise, and I am eternally grateful for the time and energy she has poured into helping me improve this project. I'd also like to thank Maziar Toosarvandani, Mia Gong, Karlos Arregi, Elena Anagnostopoulou, and John Whitman, for valuable suggestions and feedback, and audiences at WAFL18/SICOGG26, TEAL14, WCCFL43, J/K 32, and the UIUC Workshop on Korean Nominals, for all of the questions and informative discussions. Finally, a big thank you to friends and colleagues Hanyoung Byun, Jungu Kang, Seoleen Lee, Chaeun Lee, and Junho Park-Lim, for their contributions as native speakers to affirm the judgments presented in this paper, and for the many enthusiastic and insightful discussions on Korean syntax.

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